

CST8506 Assignment 2

Due: Check Brightspace for Due and End Dates

(Submissions after the end date will not be accepted)

Goal: The goal of this assignment is to work with **CIFAR10** (<https://keras.io/api/datasets/cifar10/>), and perform classification using MLP, NN, and CNN. **The main objective in this assignment is to research on various parameters and see their effect on the given dataset.**

Data Understanding & Preparation

Read about the dataset. Check results from other research findings which you can find in different resources. Print basic stats. Do the required preparation steps. Paste each code segment in the answer document along with its answers that you got when you run the code.

Modeling & Evaluation

You need to perform classification using the following methods:

1. MLP (of ScikitLearn)
2. NN with dense layers
3. CNN
 - a. Model 1: Conv layers and Dense layers as hidden layers
 - b. Model 2: Conv layers, max pool layers and Dense layers as hidden layers
 - c. Model 3: Conv layers, average pool layers and Dense layers as hidden layers
 - d. Model 4: Conv, max pool, average pool and Dense layers as hidden layers

For the CNN models, create pictorial representation of the model (check slides for reference) and a table that shows the layer name, output shape and the number of parameters for each layer.

You must show the calculation of output shape and total number of parameters in the table. **Numbers without calculation will not get any marks.** You need to create 4 pictures and 4 tables. **Tabulate all the results in one table.**

Th main part of this assignment is study about the various parameters. Discuss about the common parameters including (but not limited to):

MLP – hidden layer sizes, activation function, solver, batch size, learning rate, maximum iteration. Explain how each activation function does the calculations.

Dense layer: units, activation

Convolution layer: filters, kernel size, strides, padding, activation, input_shape

Compile: optimizer, loss, metrics

Fit: X, y, epochs, batch size, validation split, validation data

You can add more parameters as required. **You must show the effect of each option of various parameters using the given dataset and explain them in the report.** When you explain, paste the code segment and explain the effect of parameters. You must also explain the role of compile and fit methods.

Submission Details:

This is an individual assignment. Assignment should have a **cover page** with the name (Last name, first name) and student number. **Your answer document should have screenshots of every code segment along with their answers and required explanation.** You must submit your answer document AND your Colab notebook. **Missing any of these will result in a grade 0.**

Don't zip your files. Zipped files will not be graded.